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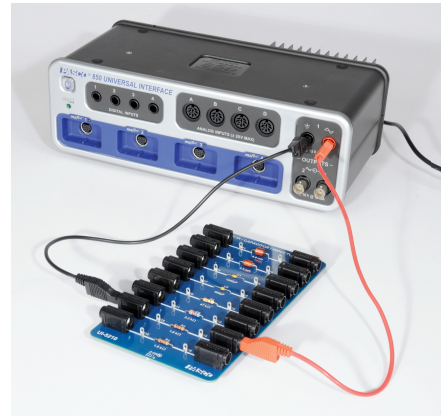
Ohm's Law

Equipment

	Included:	
1	Resistive/Capacitive/Inductive Network	UI-5210
1	Short Patch Cords (set of 8)	SE-7123
	Not Included, but required:	
1	550 Universal Interface	UI-5001

Introduction

The purpose of this experiment is to verify Ohm's Law. The equivalent resistance of series/parallel circuits is also examined. This may be performed as a single lab or two short stand-alone labs.



Written by Chuck Hunt

Theory

In metals and some other materials (in particular, commercially manufactured resistors), one finds experimentally that the voltage drop, V , across the material is directly proportional to the current, through the material (provided the temperature remains relatively constant):

$$V \propto I$$

which is referred to as Ohm's Law. It is convenient to define a proportionality constant called the resistance (unit: Ohm Ω) = V/A) such that

$$V = IR$$

$$R = V/I \quad (1)$$

A resistor generally means a device that obeys Ohm's Law (many devices do not) and has a resistance R . Two (or more) resistors can be connected in series (as in Figure 1), or in parallel (as Figure 2). Resistors could also be connected in a series/parallel circuit like Figure 3. An equivalent resistor is a single resistor that could replace a more complex circuit and produce the same total current when the same total voltage is applied. For a series circuit, the resistances are additive:

$$R_{eq} = R_1 + R_2 \quad (2)$$

where R_{eq} is the equivalent resistance. For a parallel circuit, the resistances add as reciprocals

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \quad (3)$$

A more complex circuit like Figure 3 can be handled by noting that R_1 and R_2 are in parallel and can be reduced to an equivalent resistance using Equation 3. That equivalent resistance is then in series with R_3 and can be treated using Equation 2 to find the equivalent resistance of the entire series/parallel circuit.

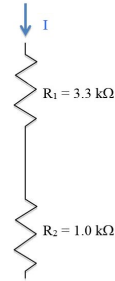


Fig. 1: Series

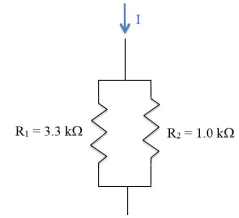


Fig. 2: Parallel

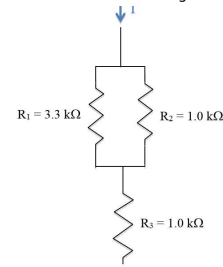


Fig 3: Series/Parallel

Pre-Lab Questions:

1. Imagine you have a simple circuit with a battery and a lightbulb connected in series, meaning the current flows from the battery, through the lightbulb, and back to the battery. Now, if you add more resistors in series to this circuit (by connecting them one after the other between the battery and the lightbulb), what happens to the brightness of the lightbulb? Does it get brighter, dimmer, or stay the same?

2. Two identical bulbs are connected in parallel. If one bulb burns out, what happens to the brightness of the remaining bulb?

3. A lightbulb operates at 12 V. If it's connected to a 6 V battery, does its brightness increase, decrease, or stay the same?

Setup/Procedure: Ohm's Law

1. Setup the circuit shown in Figure 4 using the 3.3 kΩ resistor.
2. Turn On the PASCO 550 Universal Interface.
3. Click open the Signal Generator at the left of the screen. Set 550 Output for a DC Waveform with a DC Voltage of 1V. Click On.
4. Calibration Run: Unplug the red lead from the 550 Universal Interface. The current should now be zero for all voltage, but is not. Click RECORD. Wait a few seconds (10 seconds) until the Average current reading stop drifting. Record the Average Current value in the Zero Current column of the Voltage vs Current table. If the Actual Voltage is different from the Voltage shown in the first column of the table, change the table to match the actual voltage. Click STOP.
5. Change the Signal Generator to 2V and repeat step 4. Repeat again for voltages of 3 V, 4 V, 5 V & 6 V.
6. Set the Signal Generator back to 1 V.
7. Experiment Run: plug the red lead back into the 550 Output 1 jack. Repeat steps 4 & 5 except record the values for Average current in the Measured Current column. The True Current is the difference between the Measured Current and the Zero Current.
8. Turn off the Signal Generator.

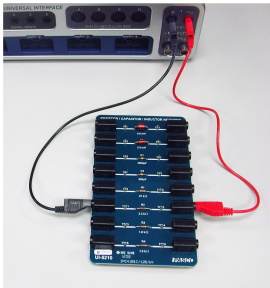


Figure 4: Ohm's Law Setup

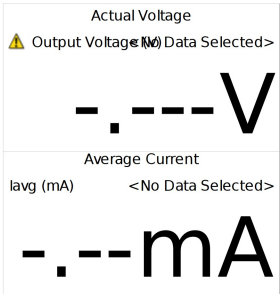


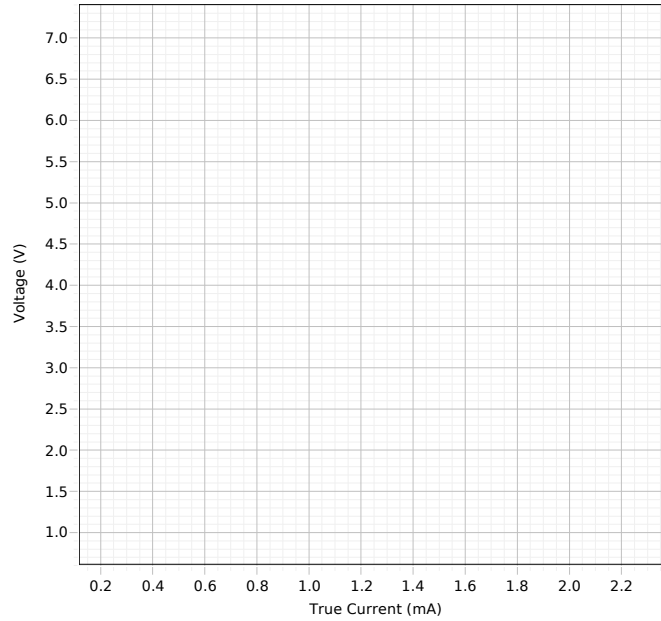
Table I: Voltage vs. Current

Voltage (V)	Zero Current (mA)	Measured Current (mA)	True Current (mA)
1			
2			
3			
4			
5			
6			
7			

Analysis: Ohm's Law

The data from the Voltage vs. Current table on the previous page is graphed on the Ohm's Law graph.

1. Click On the Scale Axes icon [the leftmost icon] on the graph toolbar to scale the axes.
2. Click on the black triangle by the Curve Fit icon on the graph toolbar and select Proportional.
3. Answer the first two Questions on the Ohm's Conclusions page.



Ohm's Law

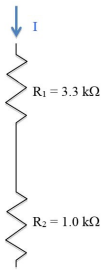


Figure 1: Series

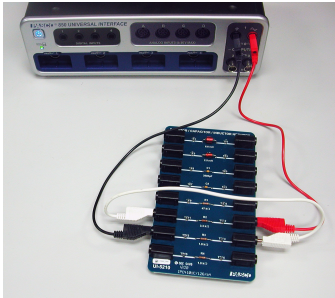


Figure 5: Series

Equivalent Circuit Diagrams:

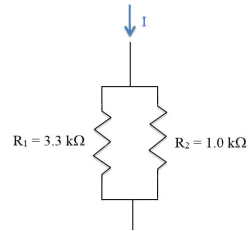


Figure 2: Parallel

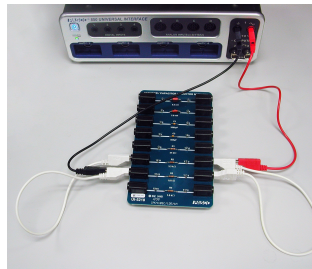


Figure 6: Parallel

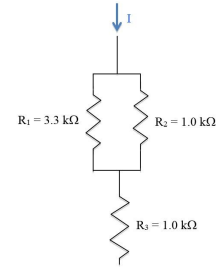


Figure 3: Series/Parallel

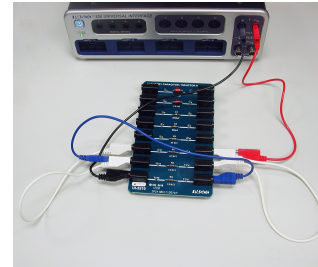


Figure 7: Series/Parallel

Equivalent Resistance

Resistor Check (Optional): The resistors on the UI-5210 circuit board are accurate to within $\pm 5\%$. This can be improved substantially if a multimeter is available. These will generally measure resistance $\pm 1\%$.

1. Disconnect all the wires from the circuit board.
2. Use a multimeter (if available) to measure the resistance of resistors R2, R3, & R4 on the circuit board. Enter the values in the Resistor Check Values table, replacing the nominal values in column 2.

Theory Resistance: Using Equivalent Circuits (see Theory tab) and the values for the resistors from Table II, calculate the equivalent resistance for each of the three circuits shown on the previous page. Enter the values in the Theory Resistance column of the Table III on this page.

Table II: Resistor Check Values

	Resistor	Resistance (Ω)
1	R1	3300
2	R2	1000
3	R3	1000
4		
5		
6		

Table III: Equivalent Resistance

	Circuit	Theory Resistance (k Ω)
1	Series	
2	Parallel	
3	Series/Parallel	
4		
5		
6		

Output 1 Voltage
⚠ Output Voltage (V) Data Selected<

-.-.-.-V

Average Current
Iavg (mA) <No Data Selected>

-.-.-mA

Table IV: Equivalent Circuits

True Cur =[Av. Current (mA)]-[Zero Cur. (mA)]							
Circuit	Output 1 Voltage (V)	Zero Cur. (mA)	Av. Current (mA)	True Cur (mA)	Exp. Resistance (kΩ)	Theory Resistance (kΩ)	% Diff (%)
Series	7.00						
Parallel	7.00						
Series/Paralle	7.00						

Procedure: Equivalent Circuits

- Click open the Signal Generator. Set 550 Output for a DC waveform with a DC Voltage of 7 V. Click On. Click the Signal Generator button to close the panel.
- Calibrate Check: With nothing attached to Output 1 of the 550 Universal Interface, click RECORD and record until the Average Current stops changing (a few seconds). Record the value of the Average Current in the Zero Cur. column of Table IV. Enter the same value in each of the three rows. If the Output Voltage reading is different from 7.00 V, enter the value in column 1, replacing the 7.00 values.
- Set up the circuit shown in Figures 1 & 5 on the previous page.
- Click RECORD and record until the Average Current stops changing. Record the value of the Average Current in the Av. Current column of Table IV. “Tr Cur” = “Av. Current” – “Zero Current”.
- Using the "True Cur" value and Equation 1 from Theory, calculate the total resistance (experimental) of the circuit. Enter the value in the Exp. Resist. column of the table.
- Set up the circuit shown in Figures 2 & 6 on the previous page. Repeat steps 4 & 5.
- Set up the circuit shown in Figures 3 & 7 on the previous page. Repeat steps 4 & 5.

Post-Lab Questions

1. How well does your data support Ohm's Law? Explain fully!

2. What is the physical meaning of the slope of the Linear Fit to the data on the Ohm's Law graph? Hint: What are the units of the slope?

3. How well does the method of equivalent circuits work?